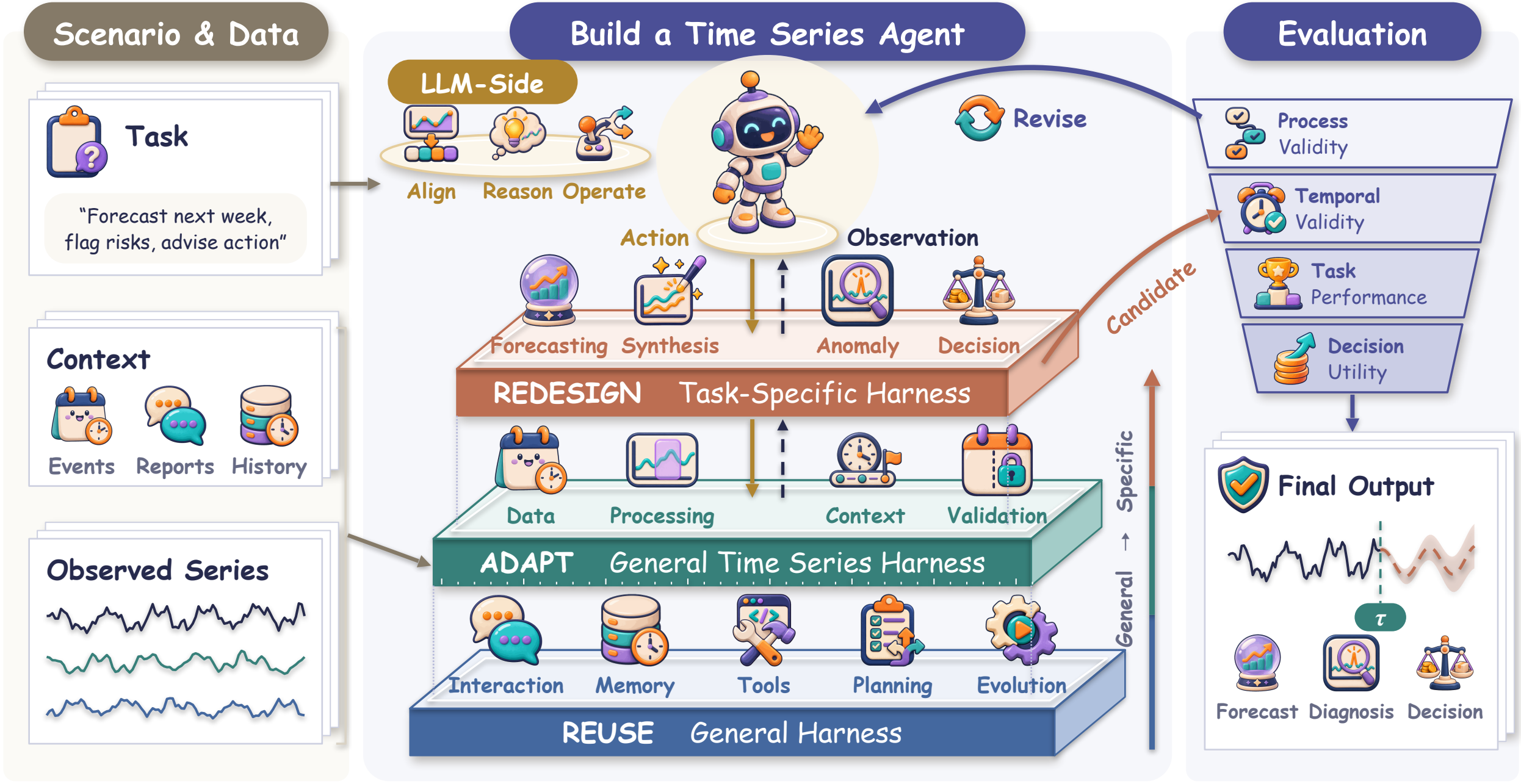
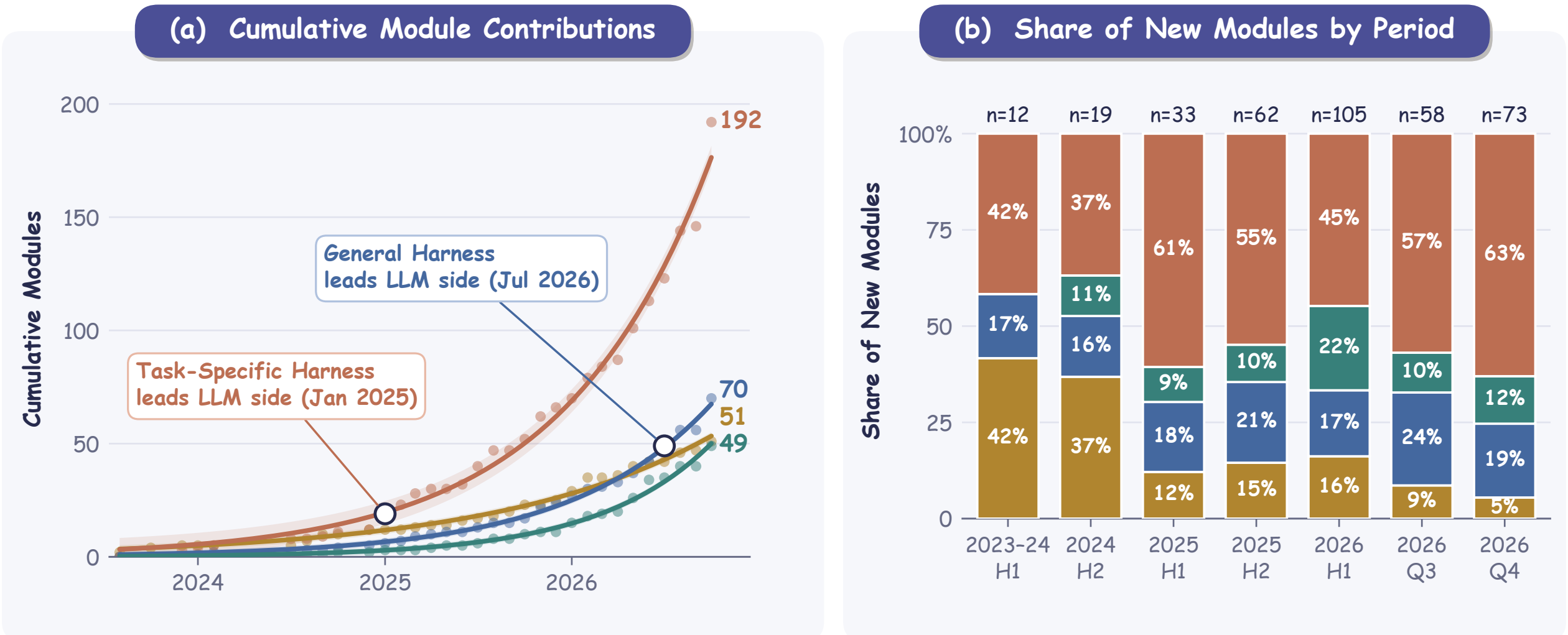

Scenario & Data
Task
"Forecast next week, flag risks, advise action"
Context
Events
Reports
History
Observed Series
Build a Time Series Agent
LLM-Side
Align
Reason
Operate
Revise
Action
Observation
Forecasting
Synthesis
Anomaly
Decision
REDESIGN
Task-Specific Harness
Data
Processing
Context
Validation
ADAPT
General Time Series Harness
Interaction
Memory
Tools
Planning
Evolution
REUSE
General Harness
Specific
General
Candidate
Evaluation
Process
Validity
Temporal
Validity
Task
Performance
Decision
Utility
Final Output
τ
Forecast
Diagnosis
Decision


LLM-Side (doubling 9.6 mo)
General Harness (doubling 6.3 mo)
General Time Series Harness (doubling 5.1 mo)
Task-Specific Harness (doubling 6.6 mo)
(a) Cumulative Module Contributions
Cumulative Modules
200
150
100
50
0
2024
2025
2026
192
70
51
49
General Harness
leads LLM side (Jul 2026)
Task-Specific Harness
leads LLM side (Jan 2025)
(b) Share of New Modules by Period
Share of New Modules
100%
75
50
25
0
n=12
n=19
n=33
n=62
n=105
n=58
n=73
42%
17%
42%
37%
11%
16%
37%
61%
9%
18%
12%
55%
10%
21%
15%
45%
22%
17%
16%
57%
10%
24%
9%
63%
12%
19%
5%
2023-24 H1
2024 H2
2025 H1
2025 H2
2026 H1
2026 Q3
2026 Q4

(a) Task Families × Shared Temporal Functions

Column: Assemble

Row: Compare

	Forecasting	Synthesis	Anomaly Diagnosis	Decision Support
Temporal Data Management	History	Samples	Streams	State
Temporal Processing	Horizon	Transforms	Intervals	Actions
Temporal Context and Memory	Events	References	Incidents	Experience
Temporal Validation	Cutoff	Fidelity	Evidence	Constraints

(b) Demand-Planning Workflow

Execute

General Harness

Prepare

General Time Series Harness

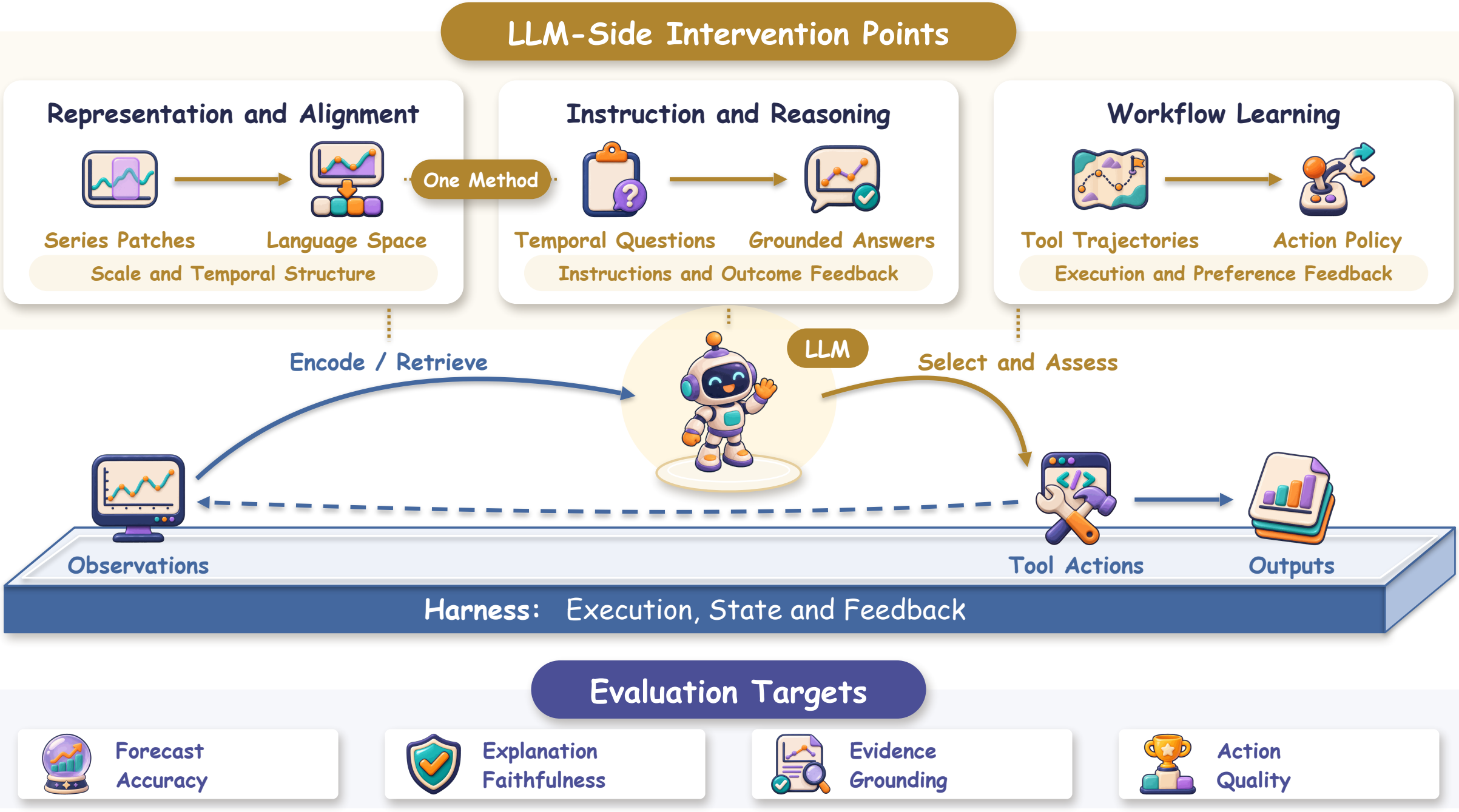
Forecast

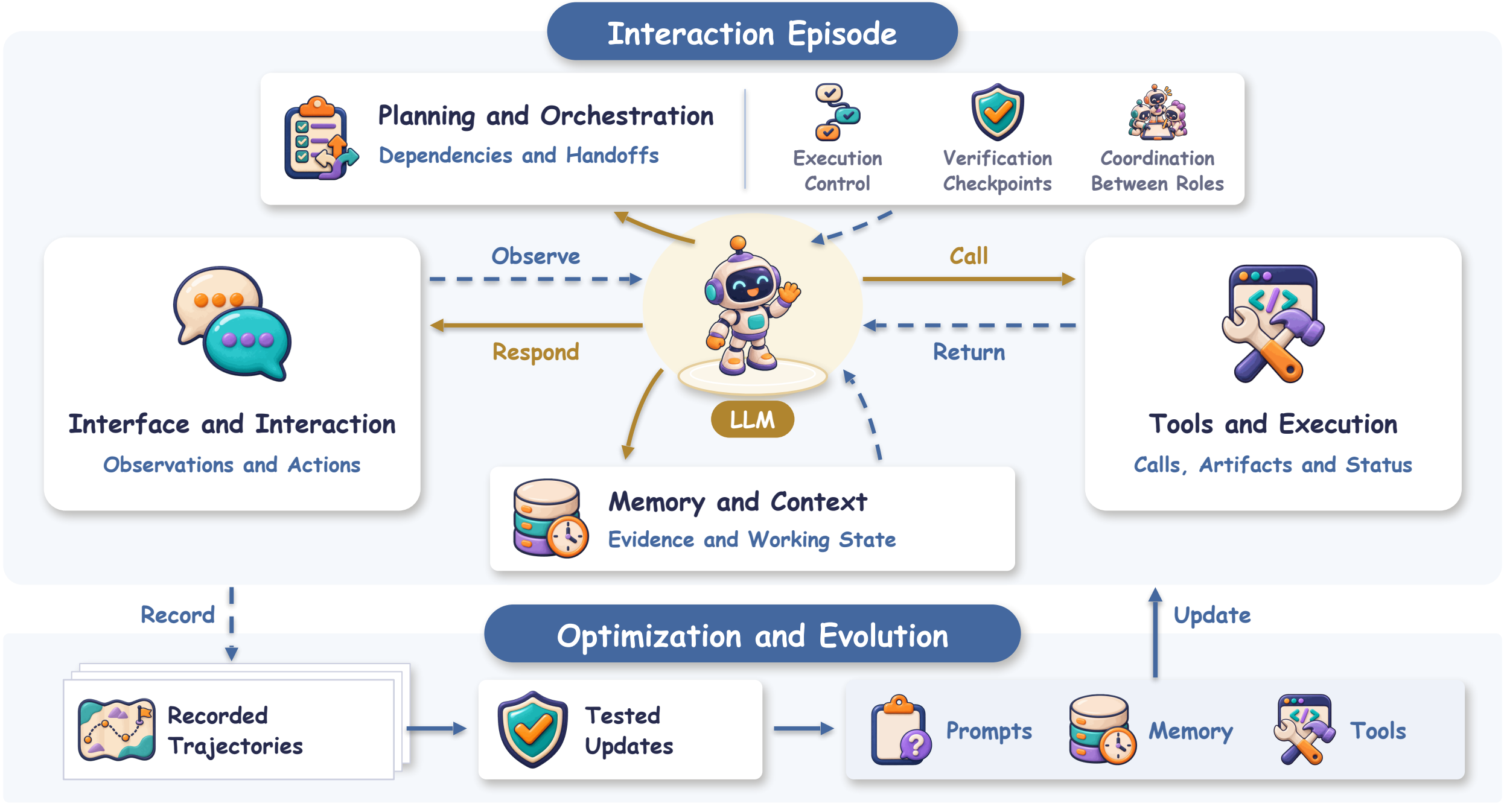
History τ Forecast

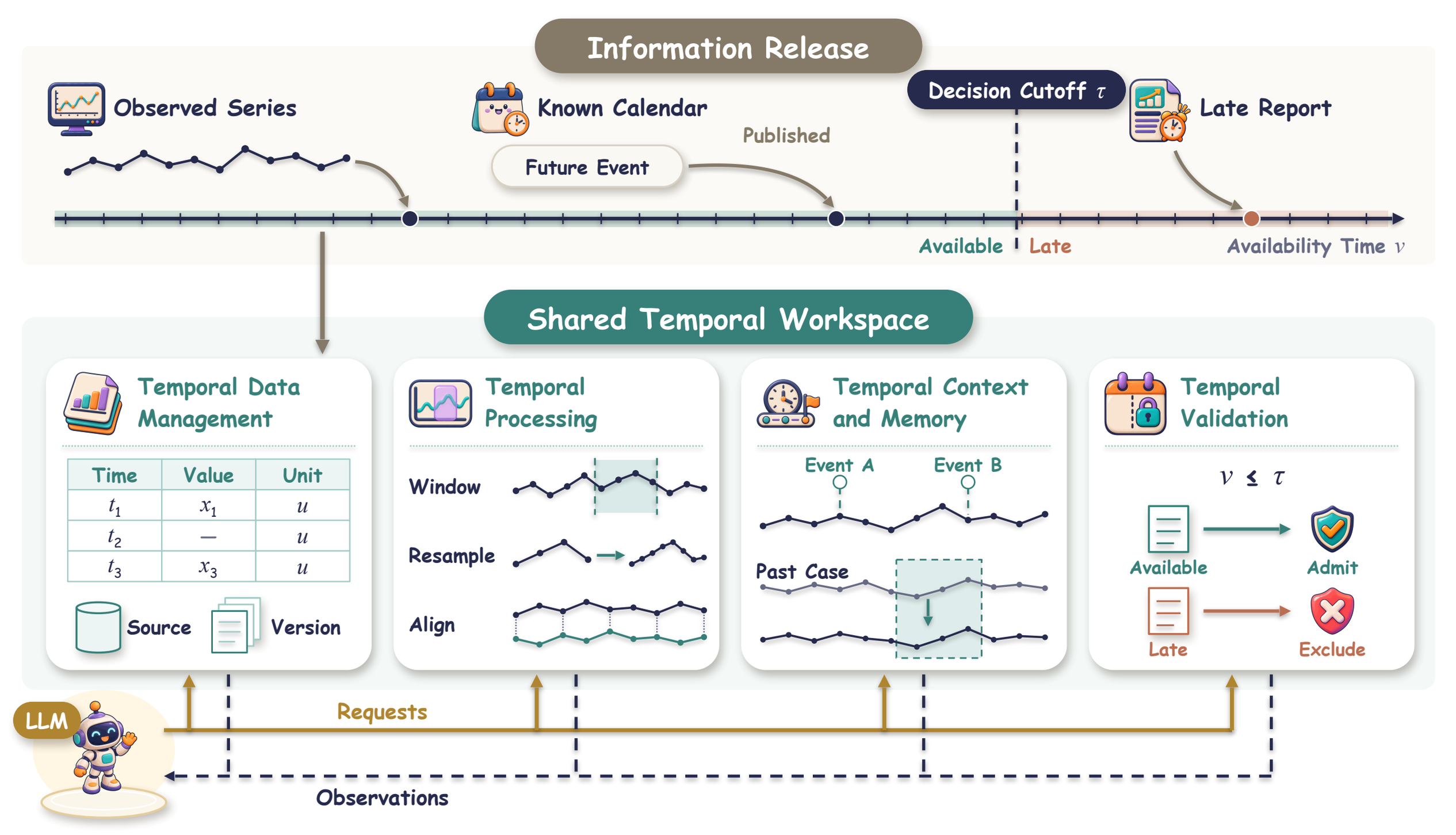
Task-Specific Harness

Decide

Task-Specific Harness

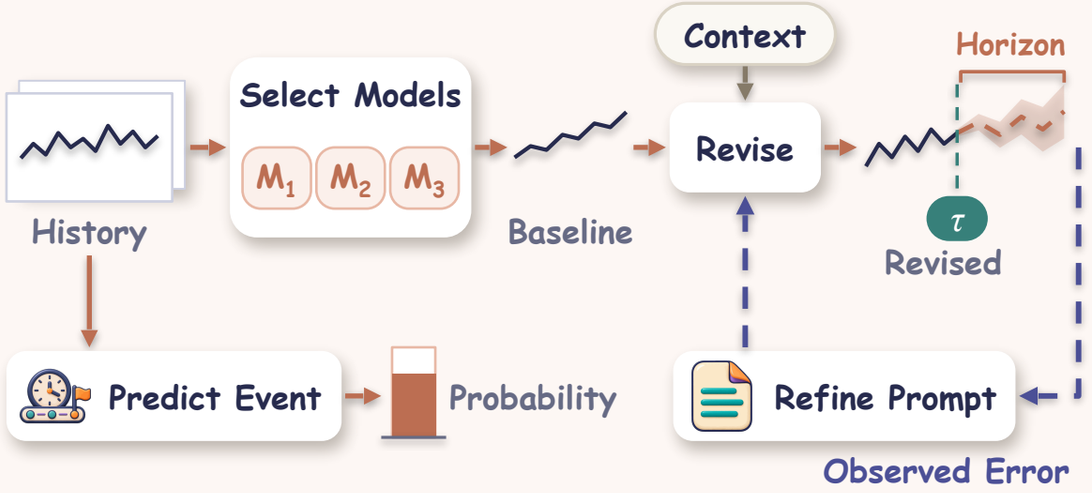

LLM-Side Intervention Points
Representation and Alignment
Series Patches
Language Space
Scale and Temporal Structure
One Method
Instruction and Reasoning
Temporal Questions
Grounded Answers
Instructions and Outcome Feedback
Workflow Learning
Tool Trajectories
Action Policy
Execution and Preference Feedback
Encode / Retrieve
LLM
Select and Assess
Observations
Tool Actions
Outputs
Harness: Execution, State and Feedback
Evaluation Targets
Forecast Accuracy
Explanation Faithfulness
Evidence Grounding
Action Quality


Interaction Episode
Planning and Orchestration
Dependencies and Handoffs
Execution Control
Verification Checkpoints
Coordination Between Roles
Observe
Respond
Call
Return
LLM
Interface and Interaction
Observations and Actions
Tools and Execution
Calls, Artifacts and Status
Memory and Context
Evidence and Working State
Record
Optimization and Evolution
Update
Recorded Trajectories
Tested Updates
Prompts
Memory
Tools


Information Release
Observed Series
Known Calendar
Decision Cutoff τ
Late Report
Published
Future Event
Available
Late
Availability Time v
Shared Temporal Workspace
Temporal Data Management
Time
Value
Unit
t_1
x_1
u
t_2
—
u
t_3
x_3
u
Source
Version
Temporal Processing
Window
Resample
Align
Temporal Context and Memory
Event A
Event B
Past Case
Temporal Validation
$v \leq \tau$
Available
Admit
Late
Exclude
LLM
Requests
Observations



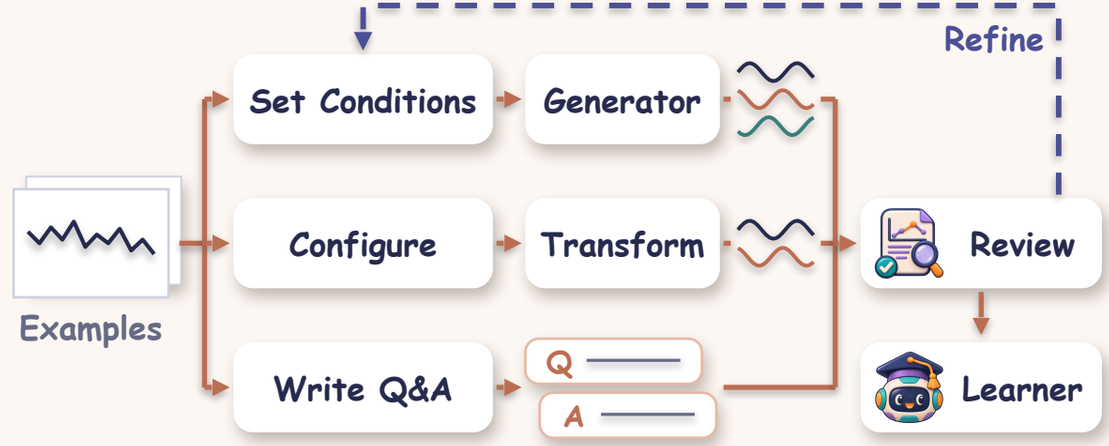
Forecasting and Prediction



Evaluation Forecast Error · Calibration



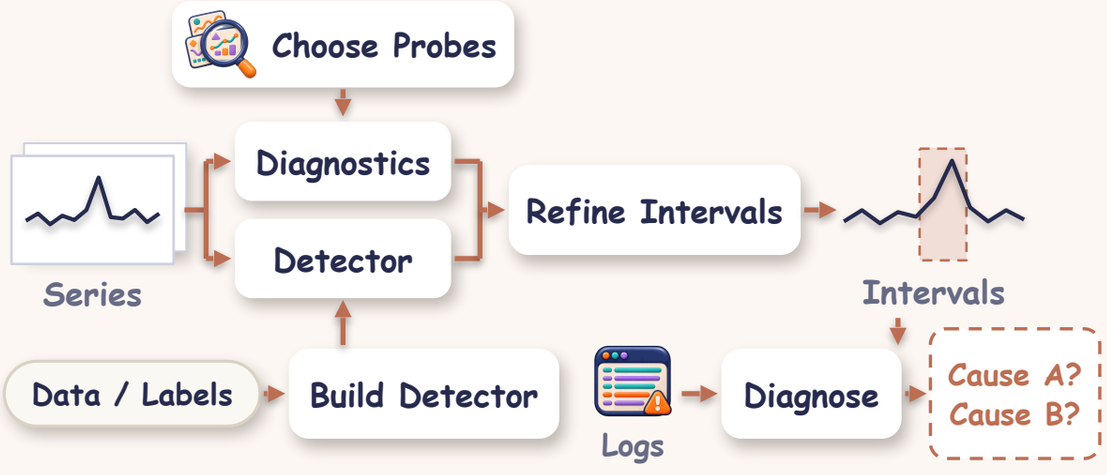
Data Augmentation and Synthesis



Evaluation Fidelity · Downstream Learning Utility



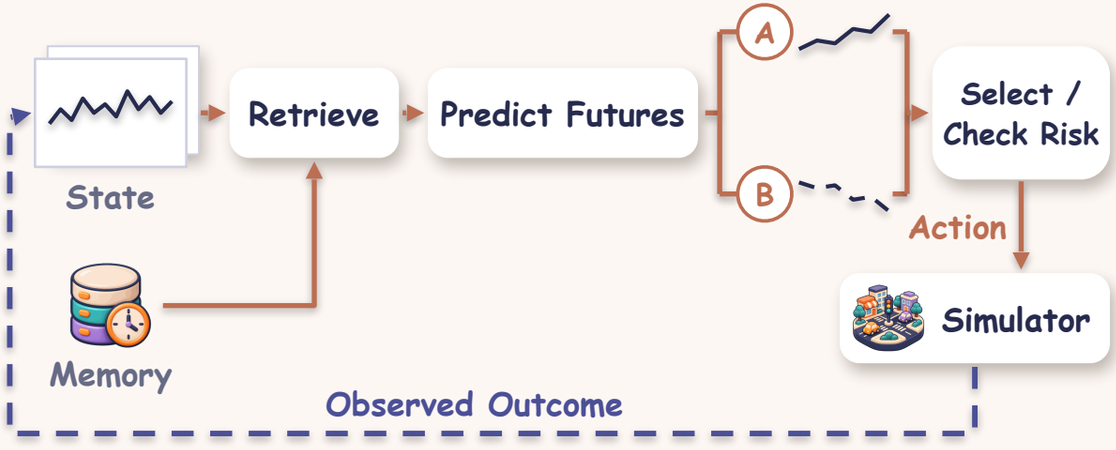
Anomaly Detection and Diagnosis



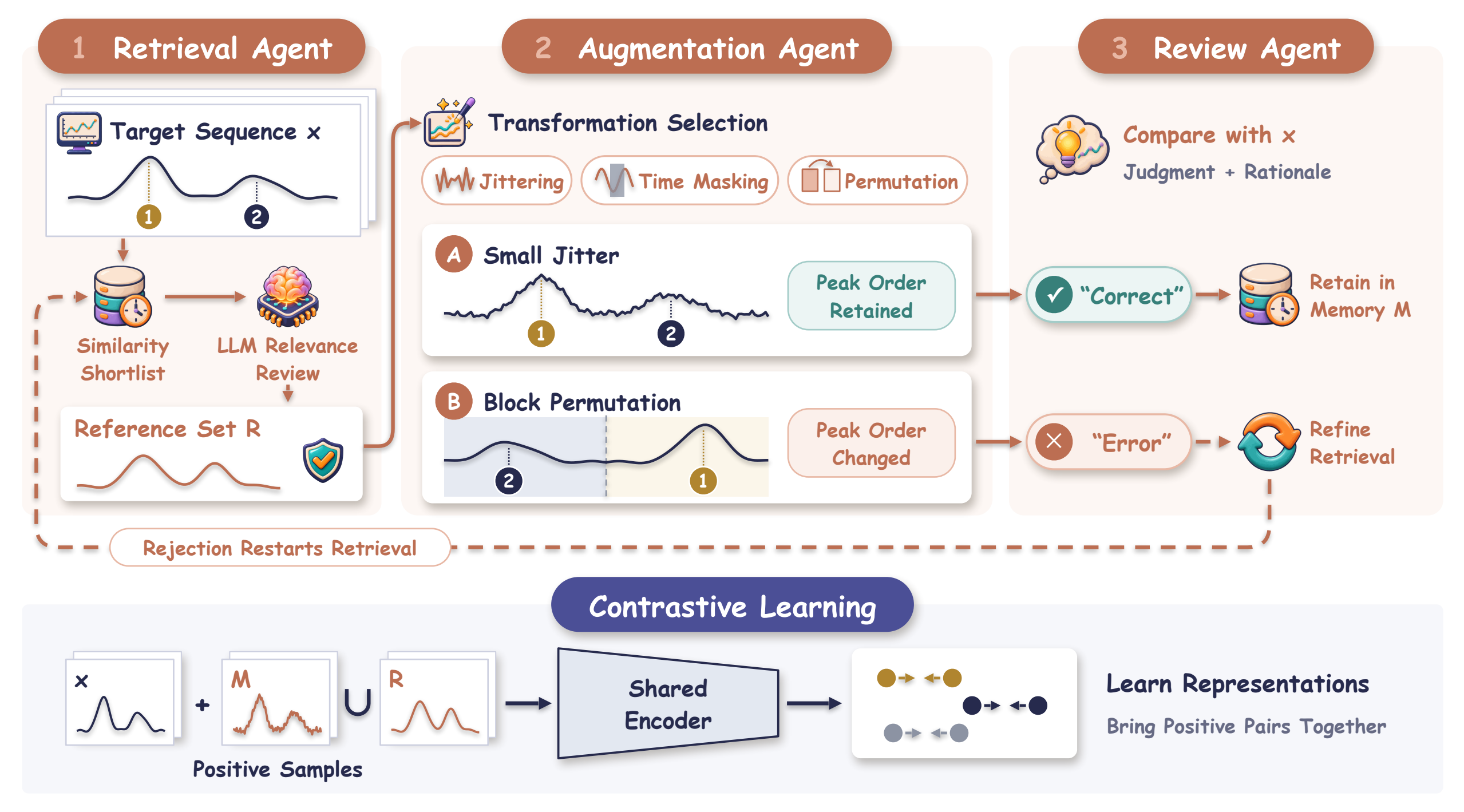
Evaluation Detection · Validation of Proposed Causes

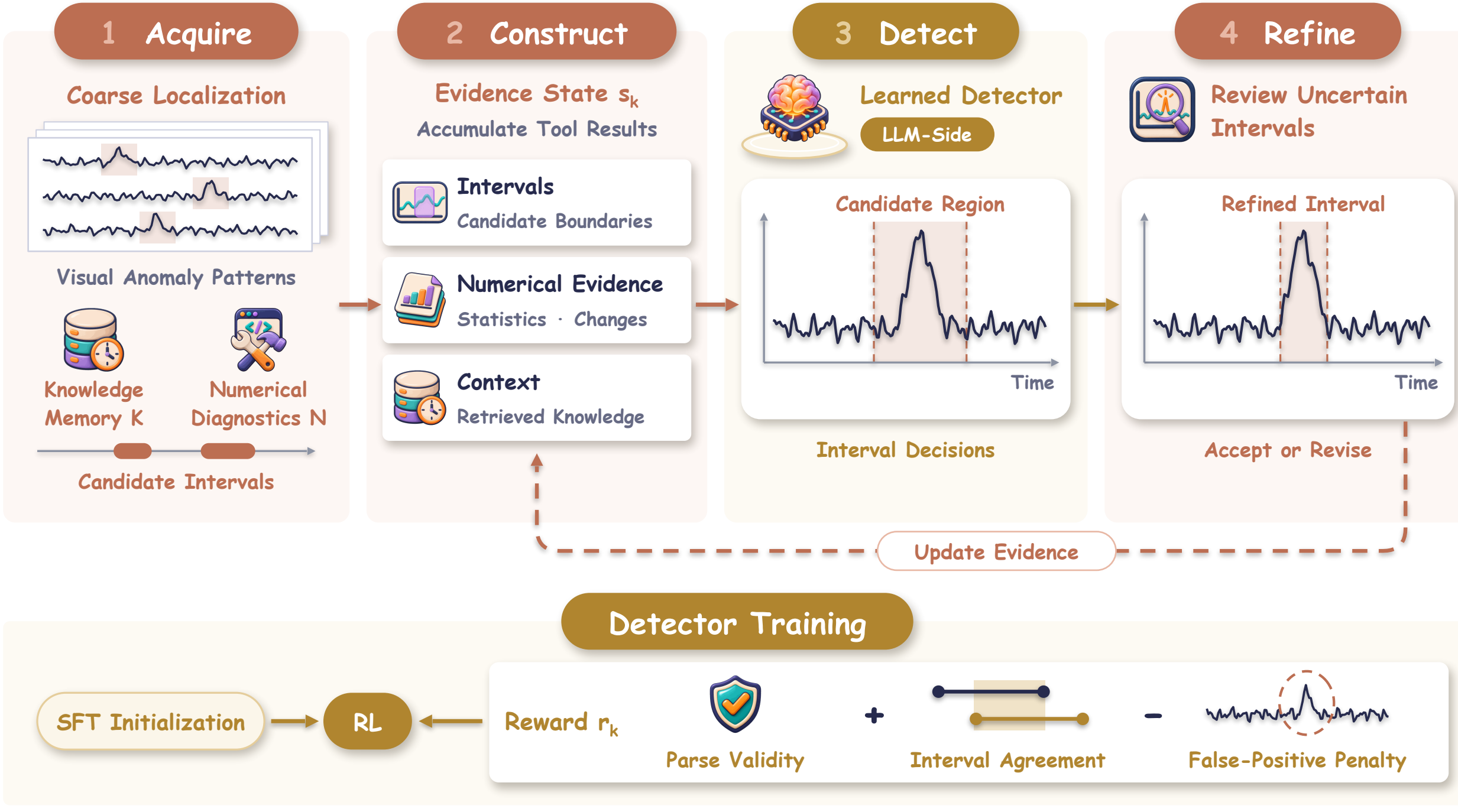


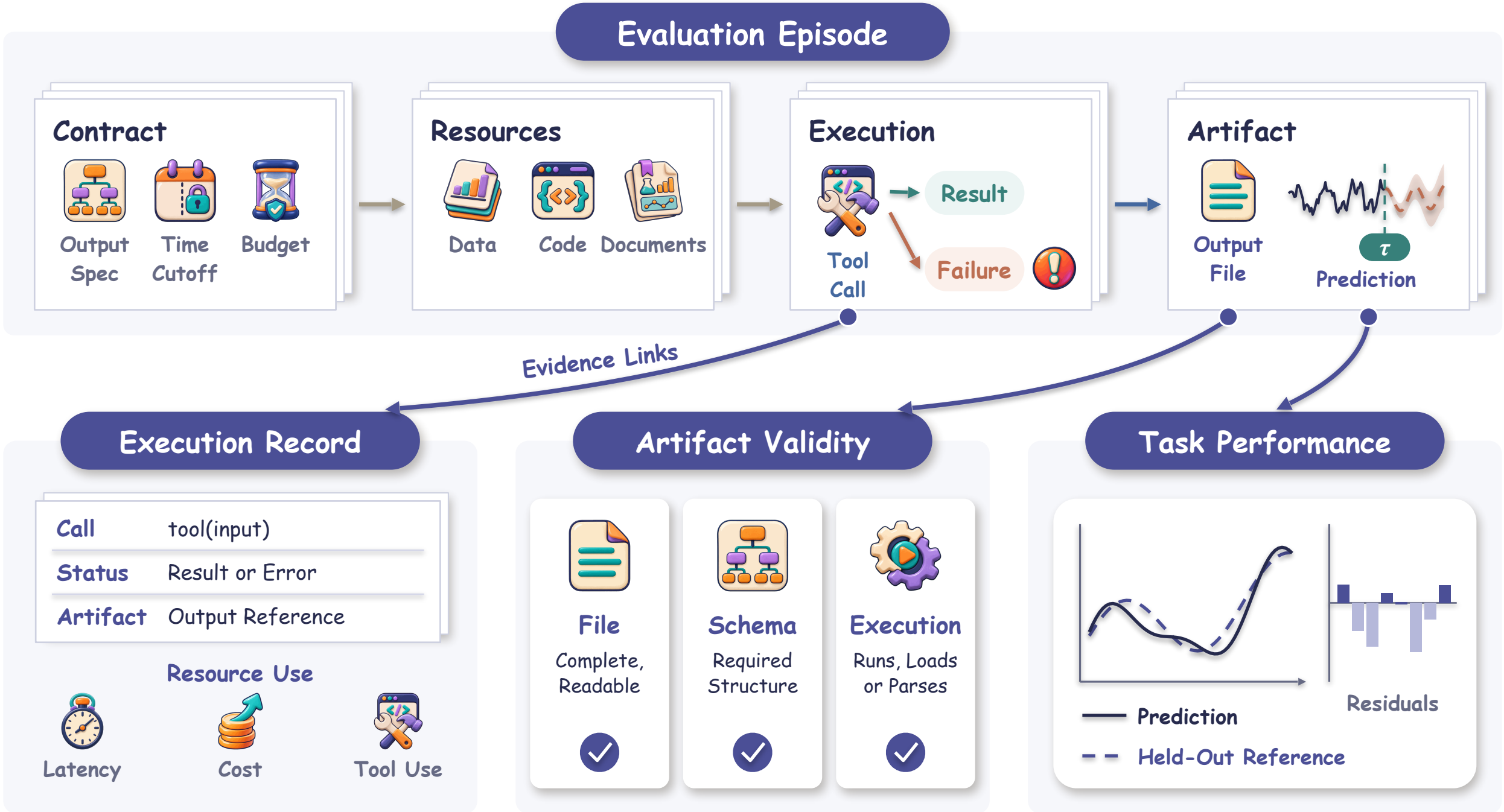
Decision Support

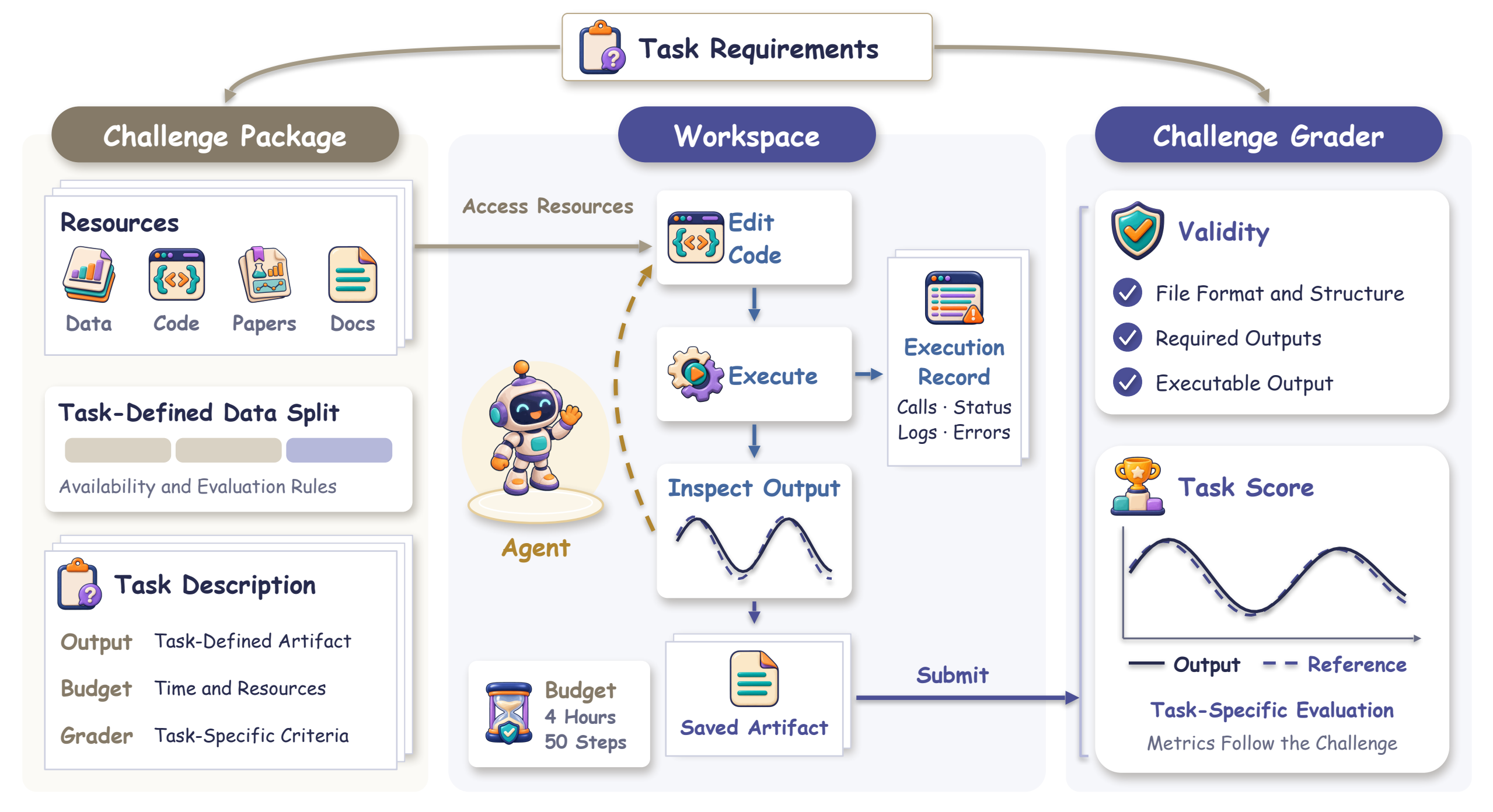


Evaluation Action Utility · Risk · Latency


1 Retrieval Agent
Target Sequence x
1
2
Similarity Shortlist
LLM Relevance Review
Reference Set R
2 Augmentation Agent
Transformation Selection
Jittering
Time Masking
Permutation
A Small Jitter
1
2
Peak Order Retained
B Block Permutation
2
1
Peak Order Changed
3 Review Agent
Compare with x
Judgment + Rationale
"Correct"
Retain in Memory M
"Error"
Refine Retrieval
Rejection Restarts Retrieval
Contrastive Learning
x
+
M
U
R
Positive Samples
Shared Encoder
Learn Representations
Bring Positive Pairs Together


1 Acquire
Coarse Localization
Visual Anomaly Patterns
Knowledge Memory K
Numerical Diagnostics N
Candidate Intervals
2 Construct
Evidence State s_k
Accumulate Tool Results
Intervals
Candidate Boundaries
Numerical Evidence
Statistics · Changes
Context
Retrieved Knowledge
3 Detect
Learned Detector
LLM-Side
Candidate Region
Time
Interval Decisions
4 Refine
Review Uncertain Intervals
Refined Interval
Time
Accept or Revise
Update Evidence
Detector Training
SFT Initialization
RL
Reward r_k
Parse Validity
Interval Agreement
False-Positive Penalty


Evaluation Episode
Contract
Output Spec
Time Cutoff
Budget
Resources
Data
Code
Documents
Execution
Tool Call
Result
Failure
Artifact
Output File
τ
Prediction
Evidence Links
Execution Record
Call
tool(input)
Status
Result or Error
Artifact
Output Reference
Resource Use
Latency
Cost
Tool Use
Artifact Validity
File
Complete, Readable
Schema
Required Structure
Execution
Runs, Loads or Parses
Task Performance
Residuals
Prediction
Held-Out Reference


Task Requirements
Challenge Package
Workspace
Challenge Grader
Resources
Data
Code
Papers
Docs
Task-Defined Data Split
Availability and Evaluation Rules
Task Description
Output
Task-Defined Artifact
Budget
Time and Resources
Grader
Task-Specific Criteria
Access Resources
Edit Code
Execute
Inspect Output
Agent
Execution Record
Calls · Status
Logs · Errors
Budget
4 Hours
50 Steps
Saved Artifact
Submit
Validity
File Format and Structure
Required Outputs
Executable Output
Task Score
Output
Reference
Task-Specific Evaluation
Metrics Follow the Challenge

Question & Series

Question

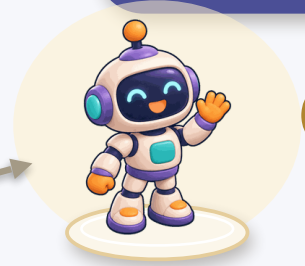
"In which part does the anomaly occur?"

Native Series

Illustrative



Iterative Analysis



Reasoner

Missing Evidence

Next Analytical Action

Detect

Presence & Interval

Observation

Evidence Log

Anomaly Found: Middle

Evidence Gate & Answer

Required Evidence

- ✓ has_anomaly = true
- ✓ anomaly_segment = middle

Both Covered

Answer: Middle

Answer Trace

Native Series → Tool + Arguments → Structured Observation → Answer Evidence

Generalist LLM

Task + Context + Observations



Select Operation & Parameters

Use the Returned Evidence

Operation + Parameters

Numerical Workspace

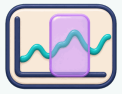
Full-Precision Series

Time	Channel 1	Channel 2
t_1	$x_{1,1}$	$x_{1,2}$
t_2	$x_{2,1}$	$x_{2,2}$

Windows & Metadata Remain Addressable

Temporal Tools

Execute Directly on the Workspace



Slice



Transform



Analyze

Compact Structured Observations

Episodic Memory



Signal Key

Shape & Statistical Structure



Context Key

Task & Textual Context



Retrieve a Successful Episode



Snowstorm → Demand

Rationale + Analytical Trace

Add to the Runtime Tool Set

Experience-Driven Tool Evolution



Executed Traces

Cluster Recurring Procedures



Candidate Tool

Code + Invocation Description



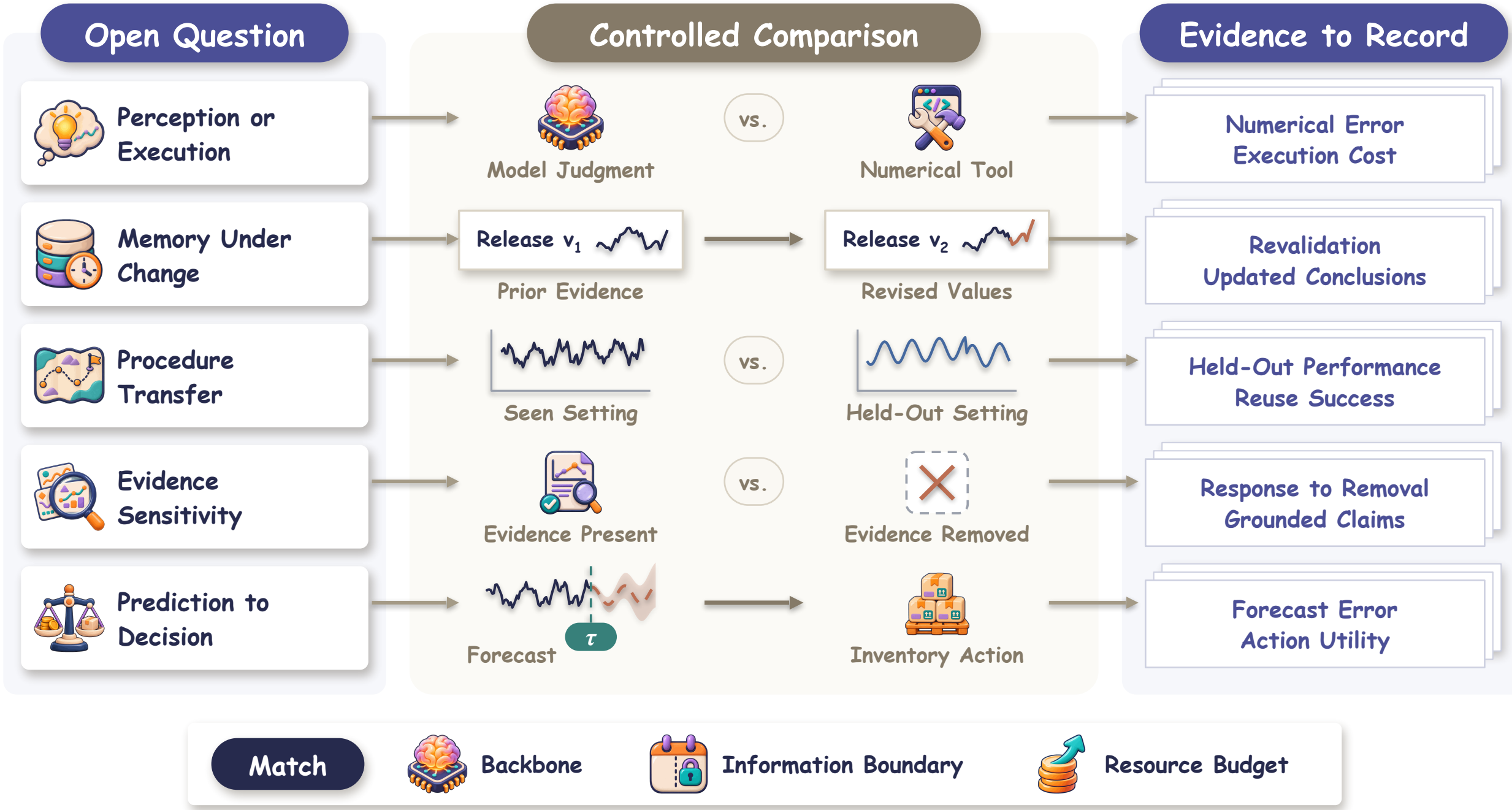
Held-Out Tasks

Meet the Success Threshold



Registered Tool

Available to Later Episodes


Open Question
Controlled Comparison
Evidence to Record
Perception or Execution
Model Judgment
vs.
Numerical Tool
Numerical Error
Execution Cost
Memory Under Change
Release v_1
Prior Evidence
Release v_2
Revised Values
Revalidation
Updated Conclusions
Procedure Transfer
Seen Setting
vs.
Held-Out Setting
Held-Out Performance
Reuse Success
Evidence Sensitivity
Evidence Present
vs.
Evidence Removed
Response to Removal
Grounded Claims
Prediction to Decision
Forecast
τ
Inventory Action
Forecast Error
Action Utility
Match
Backbone
Information Boundary
Resource Budget